

From 230V AC to Regulated 5V DC: Simulating and Designing a Linear Power Supply Using IC7805

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Project Objectives

Every electronic circuit requires a reliable DC power source, while the electrical supply available in homes is AC. Because of this, almost every electronic device contains a power supply section responsible for converting AC into a stable DC voltage.

During my electronics studies, I learned the individual concepts of transformers, rectifiers, filters and voltage regulators separately. Although I understood the theory behind each component, I wanted to see how these stages work together as a complete system.

Instead of treating this as another circuit to simulate, I decided to follow the entire energy conversion process step by step. Starting from the 230V AC mains supply, I observed how the transformer reduces the voltage, how the bridge rectifier converts AC into pulsating DC, how the filter capacitor smooths the waveform and finally how the voltage regulator produces a stable 5V output.

My this project objective was

- Design a complete regulated power supply using LTspice.
- Simulate every stage of the power supply individually.
- Understand the working principle of transformers, bridge rectifiers, filter capacitors and voltage regulators.
- Observe waveform changes at different stages of the circuit.
- Compare theoretical values with simulation results.
- Create a simulation that can later be implemented as a physical hardware project.

This approach helped me understand not only the function of each individual component but also why every stage is necessary for obtaining a reliable DC power supply.

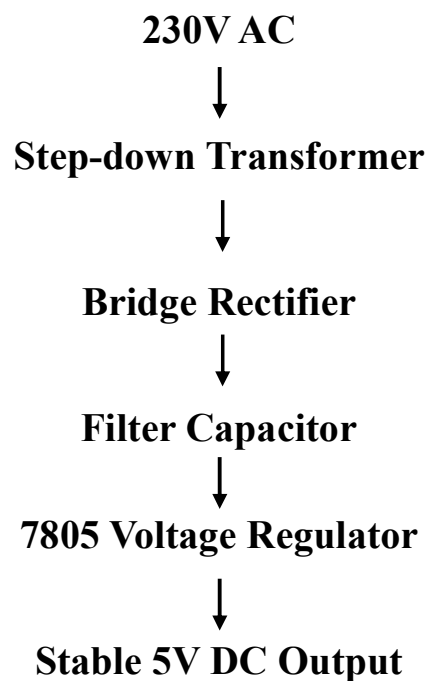
Project Specifications

Parameters	Values
Software	LTspice
Hardware	Breadboard & Components
Input Supply	230V 50Hz AC Mains
Transformer	230V/9V

Rectifier	Full Wave Bridge (1N4007)
Filter Capacitor	1000 μ F & 0.33 μ F
Regulator IC	7805
Simulation Output	5.003V
Practical Output	4.92V

Working Principle

The regulated power supply works in four main stages.



Each stage performs a specific function.

I. Step-Down Transformer

The transformer is the first stage of the circuit. Its purpose is to reduce the dangerous 230V AC mains voltage to a safer low AC voltage.

For this project,

Primary Voltage = 230V AC

Secondary Voltage = 9V AC (RMS)

The transformer works on the principle of mutual induction. When AC flows through the primary winding, it creates an alternating magnetic field inside the iron core. This changing magnetic field induces a lower voltage in the secondary winding according to the turns ratio.

The transformer also provides electrical isolation, which improves safety by separating the mains from the low-voltage circuit.

Why 9V Transformer?

A 7805 regulator requires its input voltage to be higher than 5V to regulate properly. After rectification and filtering, 9 V RMS becomes approximately:

$$V_{\text{peak}} = 9 \times \sqrt{2} = 12.727$$

After considering the voltage drop across the bridge rectifier, the capacitor charges to approximately

$$\approx 11.327\text{V}$$

This provides sufficient headroom for the 7805 regulator.

RMS and Peak Voltage

Household electricity is specified in RMS (Root Mean Square) values. However, LTspice requires the peak value when using a sine voltage source.

For example,

230 V RMS



$$230 \times \sqrt{2}$$



325.27 V Peak

Similarly,

9 V RMS



$$9 \times \sqrt{2}$$



12.73 V Peak

This is why a 325V peak source was used during simulation instead of entering 230V directly.

II. Bridge Rectifier

The transformer output is still AC, and electronic circuits require DC. Therefore, a bridge rectifier is used.

The bridge rectifier consists of four diodes connected in a bridge configuration. Its purpose is to convert both the positive and negative half-cycles of AC into pulsating DC.

Advantages:

- Utilizes both half cycles
- Higher efficiency than half-wave rectifier
- Better transformer utilization
- Lower ripple

Why 1N4007?

1N4007 is a general-purpose rectifier diode.

Reasons for selecting it:

- Maximum Reverse Voltage = 1000V
- Forward Current = 1A
- Low cost
- Easily available
- Ideal for 50Hz power supplies

Although Schottky diodes have a lower forward voltage drop, they are generally preferred in high-frequency switching power supplies (SMPS, buck converters). Since this project operates at mains frequency after the transformer, the 1N4007 is a suitable and economical choice.

III. Capacitor Filter

The output of the bridge rectifier is not perfectly smooth. It contains ripples. A large electrolytic capacitor is connected across the output. The capacitor works as an energy storage device. Its operation can be understood in two stages:

Charging

When the rectified voltage rises above the capacitor voltage, the capacitor charges rapidly.

Discharging

When the rectified voltage falls, the capacitor supplies current to the load, thereby reducing voltage fluctuations. As a result, the pulsating DC is converted into smoother DC.

Why 1000 μF ?

The ripple voltage decreases as capacitance increases. A larger capacitor stores more energy and maintains the output voltage for a longer duration between charging peaks.

Typical values:

Capacitance	Ripple
100 μF	High
220 μF	Moderate
470 μF	Low
1000 μF	Very Low

This is why I preferred 1000 μF .

Why 0.33 μF and 0.1 μF Capacitors?

Many beginners think these capacitors are optional. They are actually recommended in the 7805 datasheet.

0.33 μF (Input)

- Removes high-frequency noise coming from the input supply.

0.1 μF (Output)

- Improves regulator stability.
- Removes small high-frequency disturbances.
- Provides better transient response.

IV. 7805 Voltage Regulator

The 7805 is a fixed positive voltage regulator. It maintains a constant 5V DC output even when:

- Input voltage changes (within its operating range)
- Load current varies

Internally, it contains:

- Voltage reference
- Error amplifier
- Pass transistor
- Protection circuits

The regulator continuously compares the output voltage with its internal reference and adjusts the pass transistor to keep the output close to 5 V.

Important Specifications

Parameters	Values
Output Voltage	5V
Maximum Input Voltage	35V
Output Current	Up to 1A (with proper heatsink)
Dropout Requirement	Approximately 2V

Heat Dissipation

One limitation of linear regulators is heat generation. The power dissipated is:

$$P = (V_{in} - V_{out}) \times I_{load}$$

For example,

Input = 12V, Output = 5V & Load Current = 0.5A

Then, Power Loss = $(12 - 5) \times 0.5 = 3.5W$

This energy is converted into heat. For higher currents, a heat sink becomes necessary.

Since this project is intended for learning and testing with light loads, the regulator can often operate without a heat sink, but it must be added if the regulator becomes excessively hot during continuous operation.

LTspice Simulation, Analysis and Discussion

Why to perform Simulation first

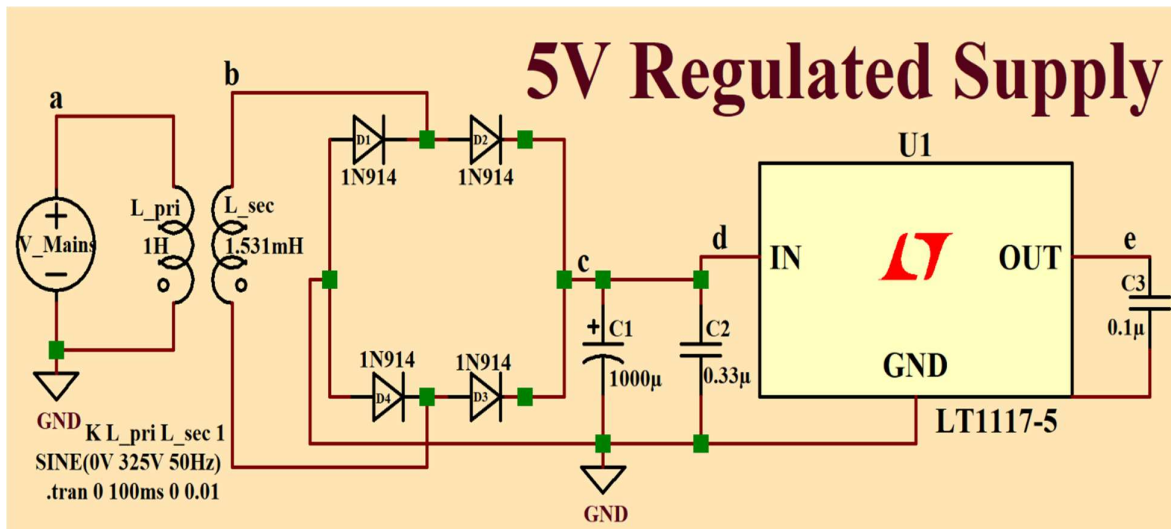
Simulation allows the behaviour of the circuit to be analysed without the risk of damaging components. It also helps verify theoretical calculations and identify design errors before physical implementation.

I simulated before hardware implementation for the following reasons:

- To verify that the circuit operates correctly before purchasing or assembling components.
- To compare theoretical calculations with simulated results.
- To observe voltage waveforms at different stages of the circuit.
- To identify wiring mistakes without damaging components.
- To understand the role of each component in the conversion process.

One of the biggest advantages of LTspice is that voltages and currents can be measured at any point in the circuit, making it an excellent learning tool

LTspice Circuit Description



Values of the components and their purpose in simulation.

Components	Values	Purpose
AC Source	325V Peak, 50Hz	Represents 230V RMS Mains Supply
Transformer	230V/9V	Steps Down Mains Voltage
Diode	$4 \times 1N914$	Bridge Rectifier
Filter Capacitor	$1000\mu\text{F}$	Ripple Reduction
Input Capacitor	$0.33\mu\text{F}$	High Frequency Bypass
Output Capacitor	$0.1\mu\text{F}$	Stability Improvement
Voltage Regulator	LT1117-5	7805 was not available

The transformer converts the mains voltage into a safer low-voltage AC signal. The bridge rectifier converts this AC into pulsating DC, which is then smoothed by the filter capacitor. Finally, the LT1117-5 voltage regulator produces a stable 5V output suitable for powering electronic circuits.

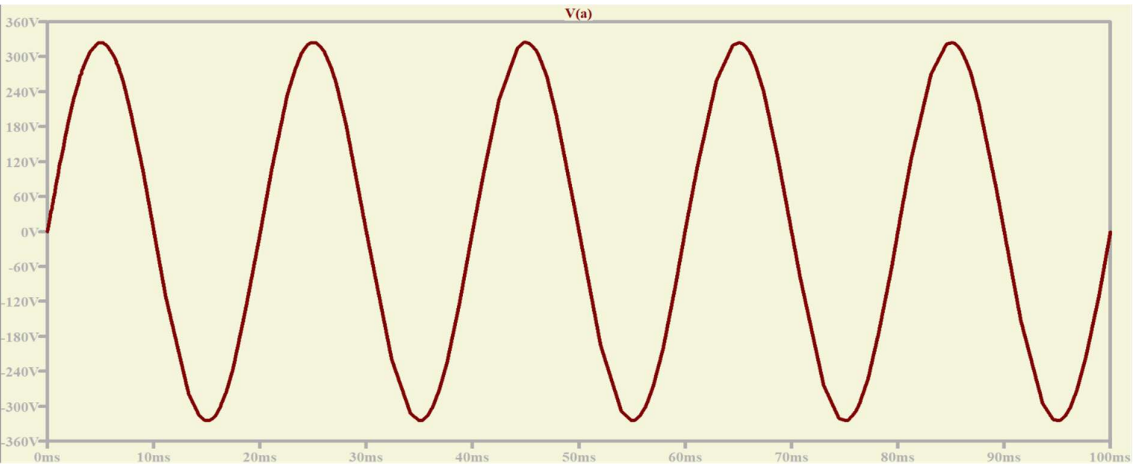
Waveform Analysis

1. Observation Point (a) – Input Supply

The first waveform I measured directly at the AC source.

Expected Waveform

- Pure sinusoidal AC.



Characteristics

- RMS Voltage = 230V
- Peak Voltage = $\pm 325\text{V}$
- Frequency = 50Hz

The waveform continuously alternates between positive and negative half cycles. This represents the actual household mains supply.

Why 325 V Was Used Instead of 230V

Electricity supplied to homes is specified in RMS voltage, whereas LTspice requires the peak value for a sine-wave source.

The relationship is:

$$V_{\text{Peak}} = V_{\text{RMS}} \times \sqrt{2}$$

Therefore,

$$230 \times 1.414 \approx 325\text{V}$$

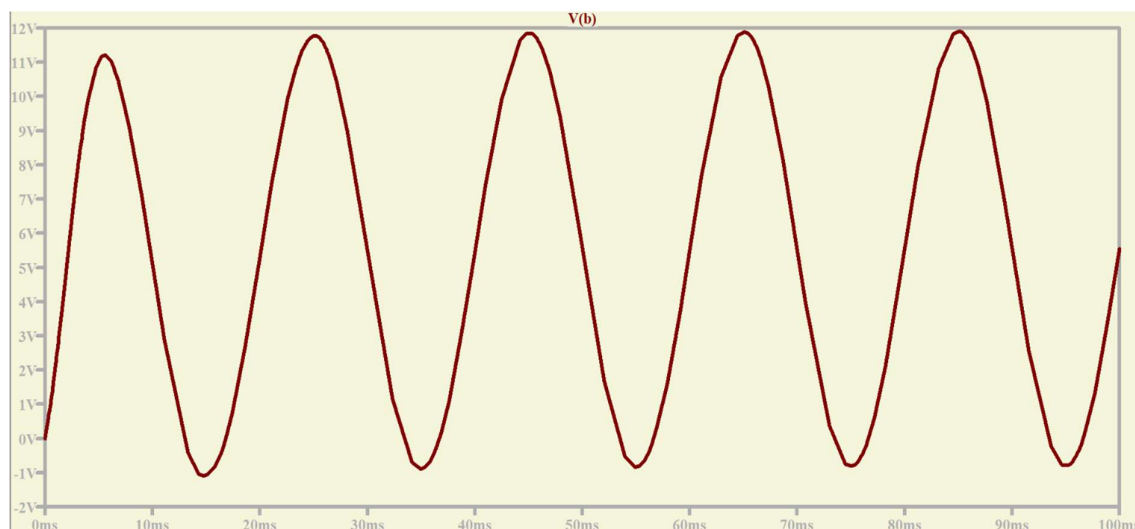
Hence, 325V Peak correctly represents 230V RMS & that's why I've used this.

2. Observation Point (b) – Transformer Output

The transformer reduces the dangerous mains voltage to a safe low-voltage AC signal.

Observed Waveform

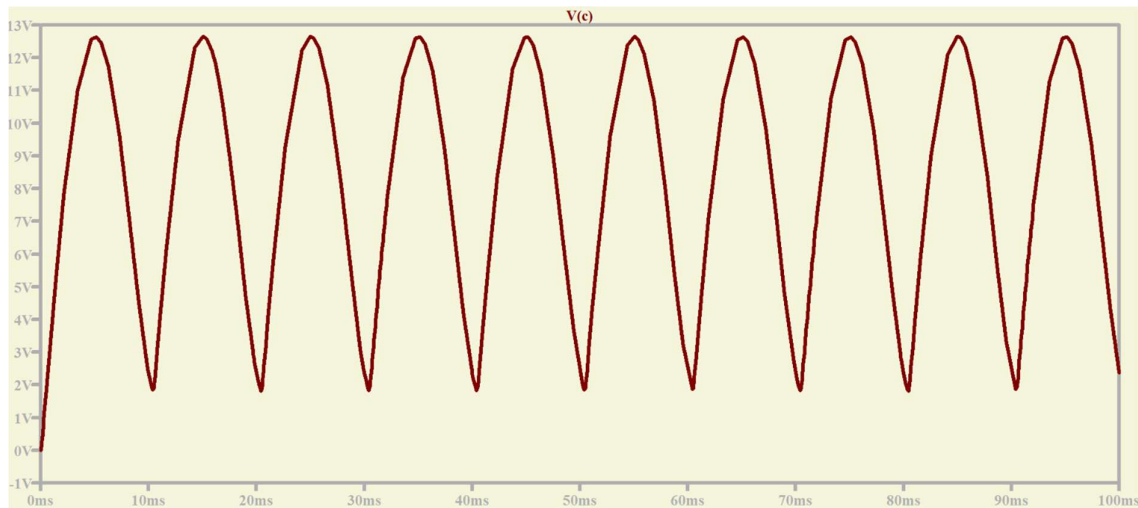
- The waveform remains sinusoidal and only the amplitude decreases.



Characteristics

- RMS Voltage $\approx 9V$
- Peak Voltage $\approx 12.73V$
- Frequency = 50Hz

3. Observation Point (c) – Bridge Rectifier Output



After passing through the bridge rectifier, the waveform changes significantly. Instead of alternating between positive and negative values, both half cycles now appear above the zero axis. This is called Full-Wave Rectification.

Observed Waveform

- Continuously Positive pulses instead of alternating between positive and negative half cycles.

Why Does This Happen?

During the positive half cycle, only two diodes conduct & during the negative half cycle, the other two diodes conduct. Although different pairs of diodes conduct, the current always flows through the load in the same direction.

Therefore, the load never experiences negative voltage. This is why the output becomes **pulsating DC**.

Why Does Frequency Become 100Hz?

The transformer output frequency is 50Hz, but the bridge rectifier uses both positive and negative half cycles. Therefore, the number of output pulses doubles.

$$\begin{aligned}\text{Output Frequency} &= 2 \times \text{Input Frequency} \\ &= 2 \times 50 = 100\text{Hz}\end{aligned}$$

This higher ripple frequency makes filtering easier because the capacitor has less time to discharge between successive peaks.

Voltage Drop across Diodes

Current passes through two diodes during each half cycle. Each silicon diode drops approximately 0.7V. Therefore,

Total voltage drop $\approx 1.4\text{V}$.

Hence, If transformer peak voltage is 12.73 V, Bridge output becomes approximately

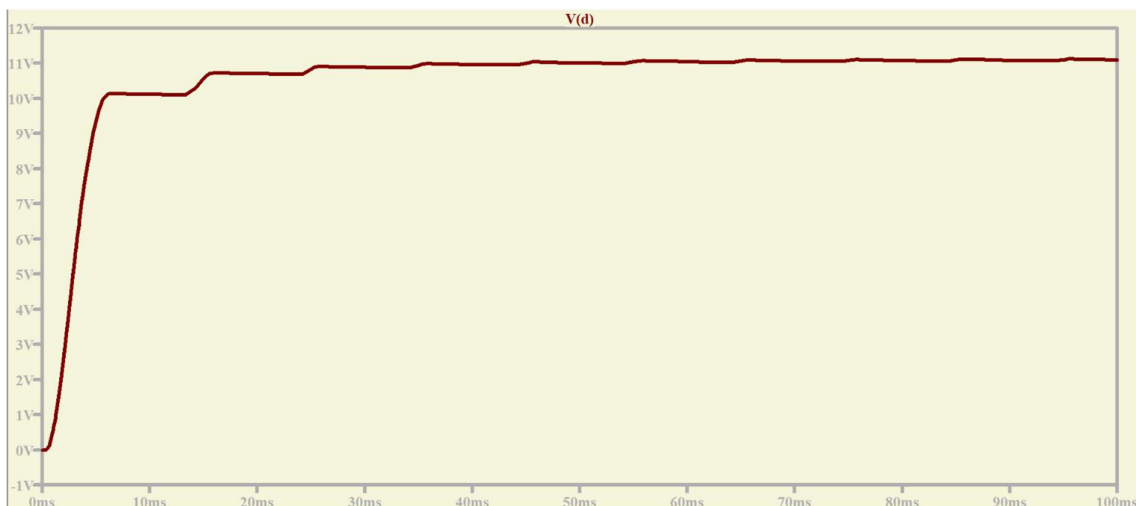
$$12.73 - 1.4 \approx 11.3\text{V}$$

This closely matches the simulation results.

Engineering Observation

Many beginners expect the bridge rectifier to produce perfectly smooth DC, but it does not. The bridge rectifier only changes the direction of current. It does not remove fluctuations. The output is still pulsating. Ripple reduction is performed by the capacitor in the next stage.

4. Observation Point (d) – After Capacitor Filter



After looking at the bridge rectifier output, one question immediately came to my mind.

If the bridge rectifier is still producing pulses, then how am I getting almost constant DC after adding just one capacitor?

At first, I thought maybe the capacitor was somehow converting AC into DC. But that's actually not what happens. The capacitor never converts AC into DC. It simply stores energy for a short time and returns it back to the circuit whenever the input voltage starts decreasing.

Let's understand it slowly.

Imagine you're filling a bucket from a tap. The tap doesn't provide water continuously. Instead, every few seconds, water comes, then stops, then comes again. If you connect the bucket directly to the tap, the water level inside the bucket doesn't suddenly become zero every time the tap stops.

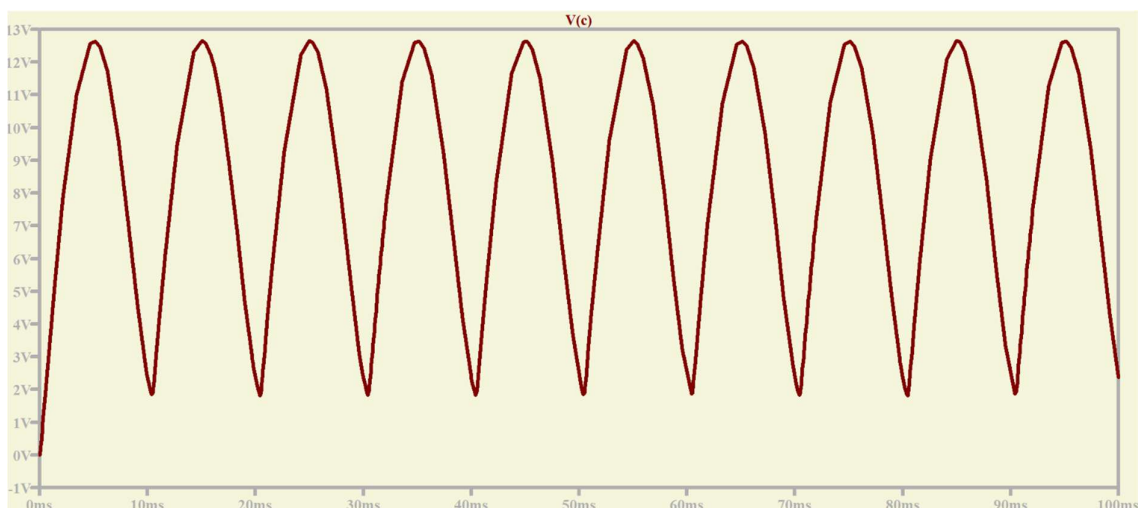
Why?

Because the bucket already contains stored water. A capacitor behaves almost the same way. Whenever the rectifier voltage increases, the capacitor charges very quickly. When the rectifier voltage starts decreasing, the capacitor doesn't immediately lose all its stored charge. Instead, it slowly supplies energy to the load.

This is why the voltage across the capacitor decreases slowly instead of falling instantly. As soon as the next peak arrives, the capacitor charges again. This charging and discharging process repeats every cycle. Because these cycles occur very rapidly (100 times every second in a full-wave rectifier), the output appears almost like a constant DC voltage.

What did I observe in LTspice?

Before adding the capacitor, the output looked like a series of hills.



There were large gaps between every peak. After adding the capacitor, those valleys almost disappeared. Instead of dropping back close to zero, the voltage remained around 11–12 V and only decreased slightly before charging again. That small up-and-down variation is called **Ripple Voltage**.

Then another question came to my mind.

If the capacitor is making the output almost constant, then why doesn't everyone simply use a huge capacitor and remove the regulator completely?

Initially I thought that using a larger capacitor would completely eliminate ripple. But that's not true. Even a 1000 μF capacitor slowly discharges while supplying current to the load. As the load current increases, the capacitor discharges faster, which increases ripple.

So although increasing capacitance reduces ripple, it can never completely eliminate it. This is one of the reasons why a voltage regulator is still required.

Why did I choose 1000 μF ?

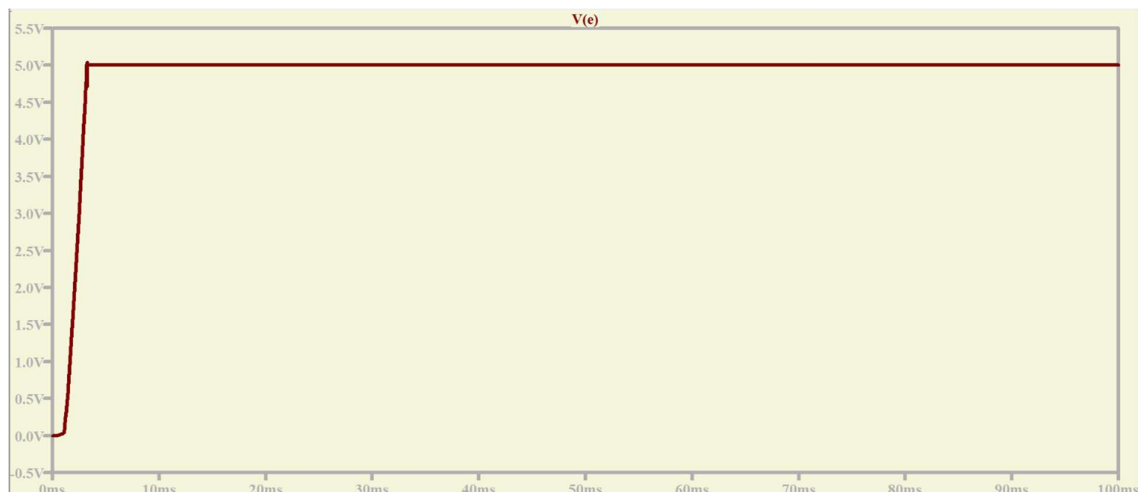
While designing this project, I had three options available.

• 100 μF	• 470 μF	• 1000 μF
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Initially, I wanted to use the biggest capacitor simply because I thought "Bigger must always be better." After studying ripple, I realized that's not always true. Larger capacitors reduce ripple, but they also occupy more space, cost more, and draw a larger charging current when power is switched on.

Since this project is a low-current educational power supply, 1000 μF provides a good balance between ripple reduction, cost and availability. If 1000 μF is not available, 470 μF can also be used, although the ripple voltage will be slightly higher.

5. Observation Point (e) – After the 7805 Voltage Regulator



After understanding how the capacitor reduces ripple, here comes the IC7805 part.

Most of the digital ICs, microcontrollers and sensors require a fixed operating voltage. Some require 5V and some 3.3V, not $\approx 11\text{V}$.

This is where the voltage regulator becomes important as we can't directly connect an Arduino to 11V. After all, the capacitor output was already around 11–12 V DC that's why we need a regulator IC.

What Actually Happens Inside the 7805?

I first imagined the regulator as a resistor that somehow drops voltage, but that explanation is incomplete. If it behaved exactly like a resistor, the output voltage would change every time the load current changed. That doesn't happen inside the 7805, there is an internal reference voltage, an error amplifier, and a pass transistor.

The regulator continuously compares the output voltage with its internal reference. If the output starts increasing, the regulator automatically increases its internal resistance (through the pass transistor). If the output starts decreasing, it allows more current to pass.

This adjustment happens continuously. Because of this feedback mechanism, the output remains very close to 5V.

My LTspice Observation

After connecting the 7805, I measured the output voltage. Instead of 4.92V or 5.24V the output remained around 5.003V. This was probably the most satisfying result of the entire simulation.

It confirmed that every previous stage

- transformer,
- bridge rectifier,
- capacitor filter,

was working correctly.

The regulator simply converted that filtered DC into a stable output suitable for electronic circuits.

One Interesting Thing I Learned

Initially, I expected the regulator to work with any input voltage. Later, I learned that it also has a limitation. The input voltage must remain sufficiently above 5V. This minimum difference is called the **dropout voltage**. For the 7805, it is approximately 2V.

Therefore, the input should normally remain above 7V, if the input falls below this value, the regulator can no longer maintain a stable 5V output. This is another reason why a 9V transformer is commonly used with a 7805 regulator.

Why Are 0.33 μF and 0.1 μF Capacitors Connected?

When I first saw the circuit, I focused only on the large 1000 μF capacitor, the two smaller capacitors looked insignificant. I even wondered, "If I already have a 1000 μF capacitor, why do I need another 0.33 μF and 0.1 μF ?"

The answer became clear after reading the datasheet. The large capacitor removes **low-frequency ripple**. However, it is not very effective against **high-frequency noise**. Small ceramic capacitors behave much better at high frequencies.

Therefore, both types of capacitors work together. The large capacitor smooths the slowly varying ripple, while the smaller capacitors remove fast transient noise. This combination improves the stability of the voltage regulator.

One important lesson I learned is that, larger capacitance does not automatically mean better performance. Different capacitor values perform different jobs.

Comparing Theory with LTspice

One thing I wanted to verify was whether the simulation actually matched the theoretical calculations. My observations are -

Parameters	Theoretical Values	LTspice Values	Observation
Input Peak Voltage	325V	325V	Same Supplied
Transformer Peak	12.73V	12.66V	Very Close

Bridge Output	11.33V	12.63V	Due to different Diode model
Regulated Output	5V	5.003V	Matches Perfectly

The small differences between theoretical and simulated values are expected because practical diode models include more realistic behaviour than ideal calculations.

One Mistake That Helped Me Learn More

While building this project, I made one mistake that actually improved my understanding. Initially, I kept measuring the voltage only at the output. Whenever I obtained an unexpected result, I had no idea which stage was responsible. That's how, I started measuring at different points –

Point 1: Input AC Mains.

Point 2: Transformer secondary coil.

Point 3: Bridge rectifier output.

Point 4: Regulator IC input voltage.

Point 5: & final regulated 5V output.

This approach immediately showed where the problem was occurring. Since then, I have realised that debugging a circuit stage-by-stage is much easier than trying to analyse the entire circuit at once. This is a habit I want to continue in future projects as well.

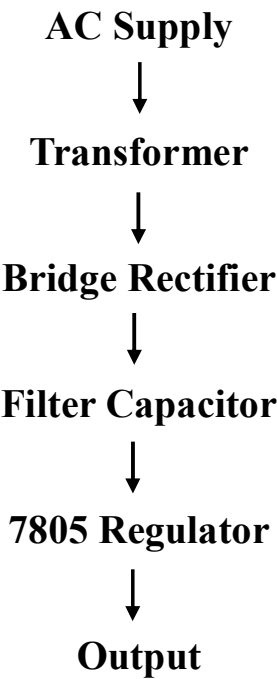
Hardware Implementation, Learning Outcomes & Future Scope

Hardware Implementation

Simulation is an excellent tool for understanding circuit operation, but practical implementation introduces many real-world factors such as component tolerances, wiring resistance, loose breadboard connections, and measurement errors.

That is why I decided to build the circuit physically after verifying the simulation.

Instead of directly assembling everything at once, I planned to follow the same sequence used during the simulation.



Building the circuit stage-by-stage made troubleshooting much easier because every section could be verified before moving to the next one.

Components Used

While selecting the components, I didn't simply choose whatever was available. I wanted to understand why each component was being used and whether there were any possible alternatives.

Components	Why I Selected?
230V/9V Transformer	Provides sufficient voltage for proper regulation while remaining commonly available.
1N4007 Diodes	Suitable for low-frequency rectification, inexpensive and capable of handling higher reverse voltage.
1000μF Electrolytic Capacitor	Provides good ripple reduction for a low-current power supply.

0.33 μ F Ceramic Capacitor (I/P)	Recommended by the 7805 datasheet to suppress high-frequency noise at the input.
0.33 μ F Ceramic Capacitor(O/P)	Improves regulator stability and filters output noise. Currently I was not having 0.1 μ F
7805 Regulator	Simple and reliable fixed 5V output regulator.
Breadboard	Allows easy modification without soldering.
Digital Multimeter	Used to verify voltages at different stages.

While collecting the components, I also realised that availability is another important part of practical electronics. Sometimes the exact component used in the simulation may not be available locally, so understanding suitable alternatives becomes equally important.

Practical Challenges

The major challenge was that selecting components is not only about electrical specifications but also about availability in my local market as I come from a small town.

For example, I wanted to keep the same capacitor values in both the simulation and the physical circuit. Since some values were not readily available, I had to compare different capacitor options and understand how changing capacitance affects ripple voltage.

These small challenges made the project much more educational than simply copying a circuit diagram.

Simulation vs Practical Circuit

Before starting this project, I assumed that simulation and hardware would produce the almost similar output.

After studying further, I realised that, this is rarely true.

Simulation assumes ideal conditions unless component models include practical characteristics. A real circuit, however, is affected by several factors.

Therefore, slight differences between simulated and practical values should always be expected. Understanding these differences is an important part of engineering rather than considering them as mistakes.

Simulation	Practical Circuit
Ideal Wiring	Wiring Resistance Present
Ideal Component Value	Component Tolerances
No Loose Connection	Loose Breadboard Contacts Possible
Constant Temperature	Temperature Affects Components
Ideal Measurements	Meter Accuracy Limitations

Comparing LTspice with Hardware Results

Parameters	Simulation	Practical Circuit	Reason Behind Differences
Input Supply	325V (Peak)	230V AC (RMS)	Mains Voltage
Voltage at Secondary	12.66V	11.1V	Due to no load behaviour of coil
Bridge Output (without filter)	12.63V	9.66V	Diode drops & Different models
Final Output	5.003V	4.92V	Ideal vs Practical Environments

What This Project Taught Me

When I started this project, I already knew the names of the main stages of a regulated power supply. I knew that a transformer, bridge rectifier, capacitor and regulator were connected one after another. However, I did not fully understand why every stage was necessary.

While working on this project, many questions came to my mind.

For example,

- Why is a bridge rectifier required?
- Why doesn't the capacitor completely remove ripple?
- Why is the regulator still required after the capacitor?

- Why are small ceramic capacitors connected even though a large electrolytic capacitor is already present?

Trying to answer these questions helped me understand the complete power supply instead of simply memorising the circuit.

Another important lesson was the value of debugging stage-by-stage.

Whenever I encountered an unexpected result, measuring every stage separately made it much easier to identify the actual problem. I believe this approach will be useful in every electronics project I build in the future.

Future Improvements

Although this project successfully produces a regulated 5V output, it also opened the door to several more advanced topics that I plan to explore in the future.

Some of the improvements I would like to work on are:

- Adjustable power supply using **LM317**
- High-efficiency switching regulators such as **LM2576** and **LM2596**
- Buck converter simulation and hardware implementation
- Boost converter
- Buck-Boost converter
- PCB design of the regulated power supply
- Efficiency comparison between linear and switching regulators
- Variable current limiting and protection circuits

This project serves as the foundation for understanding more advanced power electronic converters.

Applications

A regulated 5V power supply is one of the most commonly used circuits in electronics.

Some practical applications include:

- Arduino development boards
- CMOS logic circuits
- Digital electronics experiments

- Sensor interfacing
- Breadboard prototyping
- Educational laboratory projects

Although modern devices often use switching regulators, understanding a linear regulated power supply is essential because it introduces the basic concepts of AC to DC conversion.

Conclusion

At the beginning of this project, my goal was simply to design a regulated 5V power supply. However, while studying and simulating every stage separately, I realised that this project is much more than a simple voltage regulator circuit. It combines several important concepts of electronics, including transformers, rectifiers, capacitor filters, voltage regulators and measurement techniques.

Instead of treating the circuit as a collection of components, I tried to understand the purpose behind every stage and the reason for selecting every component. This project also improved my confidence in LTspice and showed me how valuable simulation can be before building the actual hardware.

More importantly, it changed the way I approach electronics.

- ❖ Instead of asking, "Which component should I connect next?"
- ❖ I now try to ask, "Why is this component required in the first place?"

I believe that this way of thinking will help me understand future subjects such as Power Electronics, CMOS, Embedded Systems and PCB Design much more effectively.

For me, this project is not just about obtaining a regulated 5V output. It is the first step towards building a deeper understanding of practical electronics.

References

Rather than copying information from multiple sources, I primarily used component datasheets and simulation results to understand the circuit.

Some useful references are:

- [Texas Instruments – 7805 Voltage Regulator Datasheet](#)
- Electronic Devices and Circuit Theory by Boylestad

- Sedra & Smith – Microelectronic Circuits
- Practical observations from LTspice simulation and hardware implementation